Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “work” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A **6 N** force acts at **120°** to a **5 m** displacement. **Calculate its work** and **explain the sign**.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and **record actual trials separately**.

Worked reasoning: Supplied measurements give net work **12 J** and kinetic-energy change **11 J**. Their **1 J** difference could reflect uncertainty or omitted work; **check this difference**.

Your turn: Draw two bars for these measurements and **label one check of the** experiment.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

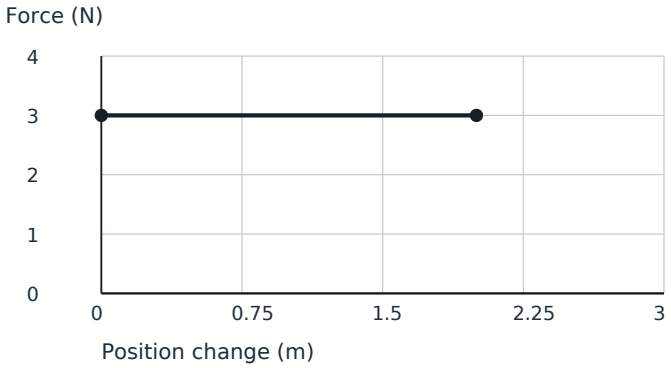
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The supplied force-displacement graph stays at **3 N** over **2 m**. **Calculate work**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **0.50 kg** lab cart speeds up from **0.40** to **1.20 m/s**. Measured pulling work is **+0.35 J**. **Complete the energy account**. Find ΔK and infer the work of resistance if every other force does zero work.

Word help: infer = use evidence to reach a conclusion

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

1. Create a trial table with mass, initial/final speed, work and ΔK.
2. **Name two causes of disagreement** that deserve checking.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **0.4 kg** projectile at **60 m/s** sticks to a stationary **3.6 kg** cart. **Find common speed** and the fraction of initial kinetic energy remaining.
Use before and after snapshots to organize the momentum account.

Word help: initial = at the beginning of the interval; subscript i

Sentence starter: I found … [unit]. This value means … in this situation.

Before **After**

Label mass, velocity and momentum.

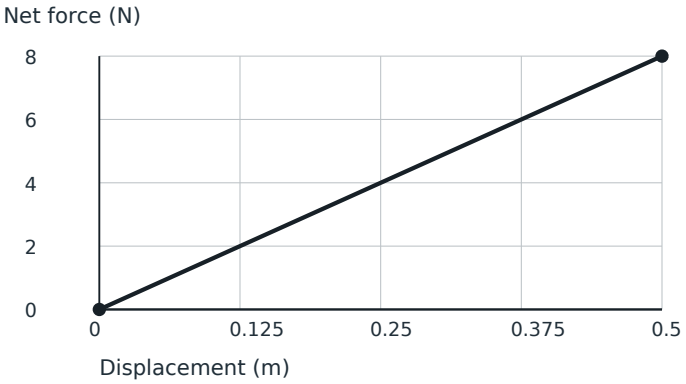
Marzano check - circle: ① ② ③

07 Exit check

A force rises linearly from **0** to **8 N** over **0.50 m**. **Find work**. If a **1 kg** cart begins at rest and this is net work, find final speed. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **2 kg** cart speeds from **1** to **3 m/s** while measured pulling work is **10 J**. Infer friction work and mean friction magnitude over **2 m** if other work is zero. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: infer = use evidence to reach a conclusion

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③